Project Design Phase: Solution Architecture

Date	19 September 2026
Team ID	
Project Name	FreelanceHub — MERN Stack Freelancing Platform
Maximum Marks	4 Marks

1. Solution Architecture Overview

Solution architecture bridges business requirements and technical implementation. Its core objectives for FreelanceHub include:

- Identifying the technical solution best suited to the identified business problem: connecting clients with skilled freelancers through secure role-based access and direct real-time communication.
- Describing the software's structure, characteristics, and behavior for anyone reviewing the project.
- Defining the development phases and specific solution requirements that guided the build.
- Establishing a clear specification for how the solution is managed and delivered.

2. Proposed Solution Architecture Diagram

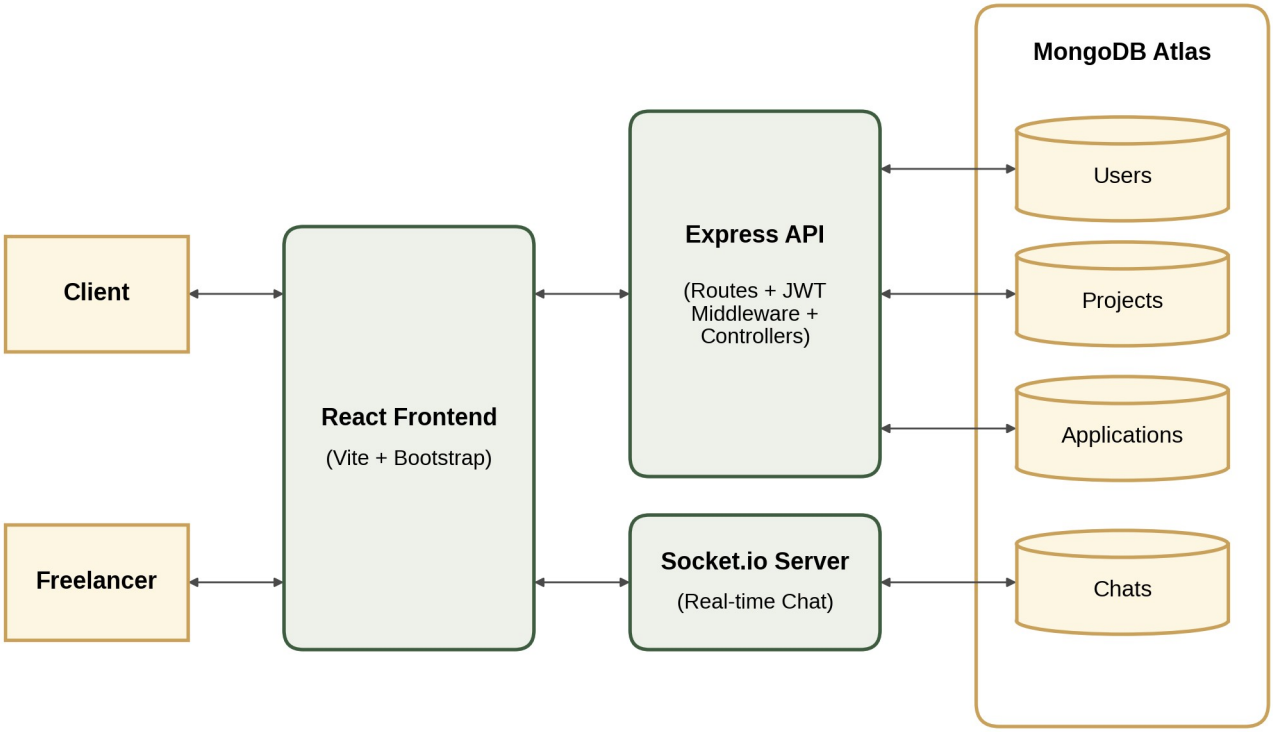


Figure 1: High-level MERN stack architecture flow for the FreelanceHub platform

Both the client and the freelancer interact with the same React frontend, which shows each of them a different dashboard based on their role. The frontend communicates with the Express API, where JWT middleware verifies every request and checks the user's role before the controllers read from or write to three core MongoDB collections — Users, Projects, and Applications — shown here as the primary data the platform depends on (Freelancer Profiles follow the same pattern). Once a client approves an application, the frontend opens a separate real-time connection to the Socket.io server, which delivers chat messages instantly between the client and the freelancer and stores them in the Chats collection.